

Inter-Area Oscillation Analysis of Power System Integrated With Virtual Synchronous Generators

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Abstract—Increasing integration of inverter-based resources (IBRs) brings new power system stability challenges. Virtual synchronous generator control-based IBRs (VSG-IBRs) are expected to take the role of some traditional SGs and reconstruct the future power system. Although VSG control imitates the main dynamics of SG, the impact of VSG-IBRs on power system stability can be different and thus worth a detailed study. In this paper, SGs in two test power systems are substituted by VSG-IBRs to investigate the impacts of VSG-IBR substitution locations, the control parameters, and the distributed or centralized substitution strategies on inter-area oscillations. The instability mechanism of inter-area oscillation as a type of low frequency oscillation (LFO) is first briefly discussed, and a selection and validation method of VSG-IBR substitution location is then proposed. The study indicates that the damping coefficient of the active power loop (D_{pVSG}) is crucial to system stability. Then, based on the proposed method, VSG-IBRs with properly designed D_{pVSG} are recommended to substitute SGs with higher participation factors (PFs) to provide more positive damping to the inter-area oscillation modes. In a large-scale system, the distributed substitution strategy contributes the positive damping of VSG-IBRs to more modes, thus further improving the system stability.

Index Terms—VSG, inter-area oscillations, stability analysis.

I. INTRODUCTION

INVERTER-BASED resources (IBRs) are key components in reconstructing future power systems with high penetration of renewables [1]. IBRs mainly adopt two types of control, i.e., grid-following and grid-forming (GFL-IBR and GFM-IBR). The GFM-IBRs have similar functions to traditional synchronous generators (SGs) and must take up a proportion to ensure the safe operation of a power electronics-dominated power system [2]. The National Renewable Energy Laboratory (NREL) of U.S. has issued a research roadmap for GFM-IBRs,

which proposes some open research questions including frequency control, voltage control, and protection as more and more traditional SGs will be substituted or retired as synchronous condensers in the future [3]. Primary research outcome shows the GFM controls can substitute/supplement governor response, voltage support, and oscillation damping functions traditionally provided by SGs [4].

Inter-area oscillation falls into the category of low frequency oscillation (LFO) according to the oscillation frequency and is a vital concern in traditional SG-based interconnected power systems [5]. The eigenvalue-based analysis is an effective method to analyze inter-area oscillations [6] by calculating the damping and oscillation frequencies of the inter-area modes and indicating how strongly SGs participate in those modes, respectively. The instability mechanism of inter-area oscillations in SG-based power systems has been well studied by previous work [7], [8], [9], and mitigation design is widely applied in real-world power systems.

In recent years, the research focus has been shifted to the impact of IBR integration on LFO [10], [11], [12], [13], [14], [15], including power oscillations among parallel connected IBRs in micro grids [10], [11], inter-area oscillations in transmission grids [12], [13], [14]. The power swing blocking with out-of-step protections affected by IBR integration is also of great interest [15]. The characteristics of power oscillations and inter-area oscillations are different. Power oscillations are mainly due to the interactions between IBRs in a close distance (i.e., parallel connection or in a small-scale test system) [10]. However, the inter-area oscillations usually involve the interactions between IBRs, and between IBRs and traditional SGs in a remote distance (i.e., connected by long transmission lines in a large-scale system) [12]. The impact of GFL-IBR on power swing blocking and out-of-step protections has been studied comprehensively in [15], while the impact of GFM-IBR needs further research. In this paper, the focus will be the impact of GFM-IBR on inter-area oscillations.

For GFL-IBRs, the phase-locked loop (PLL) dynamics are found to affect inter-area oscillations significantly [12], and the other control strategies and parameters of GFL-IBR also influence the inter-area oscillations when the penetration level increases [16]. Hence, damping controllers are designed for GFL-IBRs in PV farms, wind parks, and energy storage systems to mitigate inter-area oscillations [17], [18]. The GFM-IBR uses synchronization controls for integration rather than PLL. Hence, they have similar inter-area oscillation risks as SGs [19].

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One typical GFM-IBR is virtual synchronous generator control-based IBR (VSG-IBR), which imitates the inertia, damping, primary frequency regulation, and primary voltage regulation functions of SG [20]. However, the integration of VSG-IBRs may impact power system dynamics differently from SGs. Increasing penetration of GFM-IBRs is proved to improve the system damping under small-signal disturbances and frequency stability under large disturbances [13]. However, the study in [21] gives the opposite results: instability can also occur when the GFM-IBRs substitute a large proportion of SGs.

Different results can be caused by a lack of comprehensive analysis on multiple impact factors, such as control parameters. Inappropriate virtual moment of inertia or virtual inertia constant of VSG-IBR (J_{VSG} or H_{VSG} , respectively) [22] and the damping coefficient of the active power loop (D_{pVSG}) [23] are found detrimental to system damping. A simple “bang-bang” control is designed for VSG-IBR to provide large virtual inertia when the rotor speeds accelerate during oscillations and switch to small virtual inertia when they decelerate [24]. However, “bang-bang” control cannot consecutively change the virtual inertia. Ref. [25] proposes a control parameter-solving method to tune J_{VSG} and D_{pVSG} simultaneously to avoid the modal proximity of VSG-IBR and the rest power system. Nevertheless, this method requires complex calculations and is inconvenient to use. The transfer functions of the VSG-IBR are deducted in [26] by decoupling the active and reactive power loops into two single-input single-output (SISO) systems. Then the control parameters can be designed according to the frequency domain indicators (i.e., cross-over frequency and phase margin) [26]. The study in [27] approximates the active power loop of VSG-IBR as a first-order system to simplify the control parameter tuning process in practice. The transmission line dynamics and other converter controls are also considered in setting VSG-IBR control parameter tuning constraints to further improve the system stability [28]. For better mitigation performance, supplementary damping controllers for VSG-IBRs are also proposed to effectively damp the inter-area oscillations based on the concepts of virtual impedance [23] and power system stabilizer (PSS) [29].

Most existing research mainly investigates the impact of VSG-IBR control parameters on inter-area oscillations and control parameter tuning strategy for oscillation mitigation. There are few initiated studies of VSG-IBR substitution location (or other GFM-IBRs) impact on inter-area oscillations. In a recent study [30], two SGs in both the sending and receiving areas of a test power system are substituted by two GFM-IBRs to mitigate inter-area oscillations. The GFM-IBR placement is investigated in a small-scale system in [31]. Besides, a new inter-area mode is found between two GFM-IBRs connected by transmission lines [14]. But the substitution location impact on inter-area oscillation is not analyzed and explained clearly. Besides, the substitution strategy of GFM-IBRs in a large-scale power system can be either centralized (an HVDC station) or distributed (small-scale GFM-IBRs in several locations), which can influence the system damping differently. Hence, this paper aims to consider the impact of substitution locations and substitution strategies of GFM-IBRs on inter-area oscillations in transmission-level grids. The VSG-IBR is selected as a typical representative of GFM-IBR to conduct the study. For the other

GFM control types, similar process can be applied. The main contributions are listed below:

- 1) A selection and validation method of VSG-IBR substitution location is proposed to facilitate the inter-area oscillation analysis in large-scale and complex systems with VSG-IBR integration.
- 2) The damping characteristics and the control parameter impacts of the VSG-IBR integration considering different substitution locations and different substitution strategies are analyzed by eigenvalue-based method and validated by time-domain simulations.
- 3) Recommendations are given for VSG-IBR integration considering the coordination of substitution locations, substitution strategies and control parameters to maximize the positive effects and minimize the negative effects on the inter-area oscillation mode damping, respectively. This can serve as guidance for VSG-IBR integration in practice.

The contents of the paper are organized as follows. Section II compares the dynamic equations of VSG-IBR and SG, and then discusses the control parameter design of VSG-IBR. Section III briefly analyzes the LFO instability mechanisms and proposes a selection and validation method of VSG-IBR substitution locations. Then VSG-IBRs substitute SGs in Kundur’s two-area-four-machine (2A4M) system [6] and the New York-New England sixteen-machine sixty-eight-bus (16M68B) system [33] to investigate the above research topics in Sections IV and V, respectively. Section VI concludes the paper.

II. COMPARISON WITH SG DYNAMIC EQUATIONS AND VSG CONTROL DESIGN

This section first introduces the dynamic equations of SG and VSG-IBR. Then whether the VSG-IBR control parameters can be inherited from SG corresponding parameters is discussed by comparing their dynamic equations and working principles. A control parameter design method of VSG-IBR based on the ideas of previous papers and international standards is also proposed.

A. Dynamic Equations of SG

This paper adopts a seventh-order SG model based on [4], considering a simple excitation system, the damper and field windings of the d-axis, and two equivalent damper windings of the q-axis. The swing equations of SG are:

$$\begin{aligned} 2H_{\text{SG}} \frac{d\omega_{\text{SG}}}{dt} &= \overline{T}_m - \overline{T}_e - D_{\text{pSG}} \Delta \overline{\omega}_{\text{SG}} \\ &= \frac{\overline{P}_m - \overline{P}_e}{\overline{\omega}_{\text{mechSG}}} - D_{\text{pSG}} \Delta \overline{\omega}_{\text{SG}} \\ &\approx \overline{P}_m - \overline{P}_e - D_{\text{pSG}} \Delta \overline{\omega}_{\text{SG}} \end{aligned} \quad (1)$$

$$\frac{d\delta_{\text{SG}}}{dt} = \omega_{\text{SG}} - \omega_0 = \Delta \omega_{\text{SG}} = \omega_b \Delta \overline{\omega}_{\text{SG}} \quad (2)$$

where the variable with a cap is in per unit (pu). The change of the rotor speed ω_{SG} (in electrical radians) due to a mismatch of mechanical torque T_m and electric torque T_e is represented by (1). An assumption is made that rotor speed $\overline{\omega}_{\text{mechSG}} \approx 1$ (in mechanical radians), so that the torque and power are identical in pu. The inertia time constant H_{SG} reflects the rate at which the rotor speeds up or slows down. The term with D_{pSG} represents a speed deviation damping. The change of the rotor angle δ_{SG}

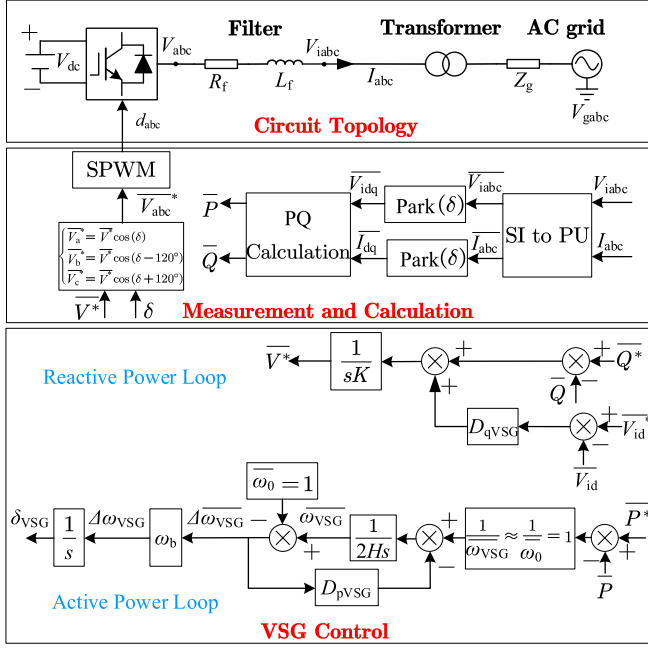


Fig. 1. Circuit and control diagrams of a typical VSG-IBR.

due to a mismatch of ω_{SG} and the rated angular speed ω_0 (in electrical radians) is represented by (2).

The simple excitation system is represented by:

$$\frac{d\bar{E}_f}{dt} = \frac{K_a (\bar{E}_t^* - \bar{E}_t) - \bar{E}_f}{T_a} \quad (3)$$

where E_f is the field voltage, E_t and E_t^* are the terminal voltage and its reference, respectively. K_a and T_a are the gain and the time constant of the excitation system, respectively. The other four state-space equations representing rotor dynamics are omitted here for simplicity, and readers may refer to [4] for details.

B. Dynamic Equations of VSG-IBR

A typical VSG-IBR is shown in Fig. 1, including an active power and a reactive power loop.

The active power loop imitates the dynamics of the SG swing equations (1) and (2) by:

$$2H_{VSG} \frac{d\bar{\omega}_{VSG}}{dt} = \frac{\bar{P}^* - \bar{P}}{\bar{\omega}_{VSG}} - D_{pVSG} \Delta\bar{\omega}_{VSG} \quad (4)$$

$$\approx \frac{\bar{P}^* - \bar{P}}{\bar{P}^* - \bar{P} - D_{pVSG} \Delta\bar{\omega}_{VSG}}$$

$$\frac{d\delta_{VSG}}{dt} = \omega_{VSG} - \omega_0 = \Delta\omega_{VSG} = \omega_b \Delta\bar{\omega}_{VSG} \quad (5)$$

Similarly, ω_{VSG} , δ_{VSG} , H_{VSG} and D_{pVSG} are the virtual rotor angular speed (in electrical radians), virtual rotor angle, virtual inertia time constant, and the damping coefficient of active power loop.

The reactive power loop imitates the dynamics of the SG excitation system (3):

$$\frac{d\bar{V}^*}{dt} = \frac{D_{qVSG} (\bar{V}_{id}^* - \bar{V}_{id}) + \bar{Q}^* - \bar{Q}}{K} \quad (6)$$

where V^* is the modulation voltage magnitude, V_{id} and V_{id}^* are the d-axis terminal voltage and its reference, respectively. K is the constant gain, and D_{qVSG} is the damping coefficient of the reactive power loop to regulate voltage.

The L-type filter dynamics determine the currents of the VSG-IBR:

$$\bar{V}_{dVSG} + \bar{\omega}_n \bar{L}_f \bar{I}_{qVSG} - \bar{V}_{idVSG} = \bar{R}_f \bar{I}_{dVSG} + \frac{\bar{L}_f}{\omega_b} \frac{d\bar{I}_{dVSG}}{dt} \quad (7)$$

$$\bar{V}_{qVSG} - \bar{\omega}_n \bar{L}_f \bar{I}_{dVSG} - \bar{V}_{iqVSG} = \bar{R}_f \bar{I}_{qVSG} + \frac{\bar{L}_f}{\omega_b} \frac{d\bar{I}_{qVSG}}{dt} \quad (8)$$

Therefore, five state-space equations of VSG-IBR are (4)–(8) considering the dynamics of ω_{VSG} , δ_{VSG} , V^* , I_{dVSG} and I_{qVSG} .

C. Discussions on VSG-IBR Control Parameter Design

1) *Active Power Loop Parameters:* VSG-IBR active power loop equations (4) and (5) are nearly the same as SG swing equations (1) and (2). There is also a model simplification that the virtual rotor speed $\bar{\omega}_{VSG} \approx \bar{\omega}_0 = 1$ (in electrical radians). The difference between the two assumptions lies in the angular speed, which is in electrical radians for VSG-IBR and in mechanical radians for SG. The original torque definition of SG is the division of power by mechanical angular speed. Therefore, the virtual torque definition of VSG-IBR needs to be modified into the division of power by electrical angular speed, as indicated in existing work [34]. Under this assumption, the term $\bar{P}^* - \bar{P}$ in VSG-IBR matches the term $\bar{P}_m - \bar{P}_e$ in SG. Hence, VSG-IBR could inherit SG parameters H_{SG} and D_{pSG} as its H_{VSG} and D_{pVSG} values under the same power ratings to approximate the dynamics of SG. The control parameters of VSG-IBR may be tuned in a wide range without the physical limitations of SG to boost some functions, like providing inertia and damping to the system. The detailed parameter design procedure of the active power loop is discussed below:

According to the diagram of Fig. 1, the damping coefficient of the active power loop D_{pVSG} can be written as:

$$D_{pVSG} = \frac{\bar{P}^* - \bar{P}}{\Delta\bar{\omega}_{VSG}} = \frac{\Delta\bar{P}}{\Delta\bar{\omega}_{VSG}} \quad (9)$$

Then D_{pVSG} may be designed as a droop coefficient based on the ability of VSG-IBR to regulate active power output under the disturbance of frequency change. The EU standard EN 50438 [35] recommends a 100%P/Hz - 16.7%P/Hz (equivalent to 1.6% - 10% droop) active power response to frequency change for IBR integration. The design of “P- ω ” droop for IBR can have a wider range, which is more flexible compared to droop function in governor control of traditional SG, with typical values of 4% - 5% droop [4]. In this paper, the highest 100%P/Hz criterion is chosen to design D_{pVSG} :

$$D_{pVSG} = \frac{\Delta\bar{P}}{\Delta\bar{\omega}_{VSG}} = \frac{1 \text{ pu}}{1 \text{ Hz}/60 \text{ Hz}} = 60 \quad (10)$$

which means a 1 Hz grid frequency change leads to 100% active power output. Normally, D_{pSG} is not very large due to physical limitations of SGs, so the droop function is realized in governor control. However, D_{pVSG} can be designed to boost the active

power response to frequency change. In Section IV-C, D_{pVSG} will also be proved crucial to the inter-area mode damping.

Similar to the time constant of SG, $2H_{\text{VSG}}$ can be designed as the time required for rated power to accelerate the virtual rotor angular velocity ω_{VSG} from 0 to rated speed ω_0 . In this paper, H_{VSG} inherits the value of H_{SG} to keep the inertia characteristics constant. However, the difference between virtual inertia in VSG-IBR and real inertia in SG needs to be investigated further in future work. The VSG-IBR may also be designed as an inertia source for a low-inertia regional power system. The design method of H_{VSG} can be found in [26] by setting cross-over frequency and phase margin if the VSG-IBRs do not need to approximate the inertia characteristics of SGs.

2) *Reactive Power Loop Parameters*: Although the VSG-IBR reactive power loop equation is similar to the excitation system equation of SG, the mechanism of regulating terminal voltage is different. The SG terminal voltage magnitude mismatch $\overline{E}_t^* - \overline{E}_t$ is regulated by controlling the field voltage E_f in (3). Then the field voltage E_f changes the generator terminal voltage V_{dSG} and V_{qSG} through rotor dynamics. However, the VSG-IBR terminal voltage mismatch $\overline{V}_{\text{id}}^* - \overline{V}_{\text{id}}$ is regulated by controlling the VSG-IBR modulation voltage V^* in (6). Then the VSG-IBR modulation voltage V^* will become the VSG-IBR output voltage magnitude V_{abc} through sinusoidal pulse width modulation (SPWM), influencing terminal voltage $\overline{V}_{\text{iabc}}$ by filter dynamics (7) and (8).

Besides, the VSG-IBR reactive power loop has reactive power – voltage (Q – V) droop, corresponding to the droop load compensation of modern SG excitation systems that ensures safe operation of parallel connected SGs. To conclude, the modelling of VSG-IBR does not consider the dynamics of SG rotor windings, and the droop load compensation of SG excitation system is not represented by the gain and time constant. Hence, the VSG-IBR parameters D_{qVSG} and K should not be inherited from SG excitation system parameters K_a and T_a . Moreover, the ignorance of SG rotor winding dynamics may also influence damping characteristics. However, the reactive power loop parameters will be later verified to have a minor influence compared to active power loop parameters on inter-area modes in Section-B. The detailed parameter design procedure of the reactive power loop is discussed below:

The damping coefficient of the reactive power loop D_{qVSG} can be written below according to Fig. 1:

$$D_{\text{qVSG}} = \frac{\overline{Q}^* - \overline{Q}}{\overline{V}_{\text{id}}^* - \overline{V}_{\text{id}}} = \frac{\Delta \overline{Q}}{\Delta \overline{V}_{\text{id}}} \quad (11)$$

Then D_{qVSG} may be designed as a droop coefficient based on the ability of VSG-IBR to regulate reactive power output under the disturbance of voltage magnitude change. The FERC order No. 827 [36] recommends a dynamic reactive power within the power factor range of ± 0.95 and gives an example of a 33%Q/3%V (9% droop) curve. The design of Q-V droop for VSG-IBR is also different from the reactive droop compensation of SG exciter, with typical value of 2% droop [4]. In normal operation, the control priority of VSG-IBR is given to active power, so the reactive power output capability is limited. Hence,

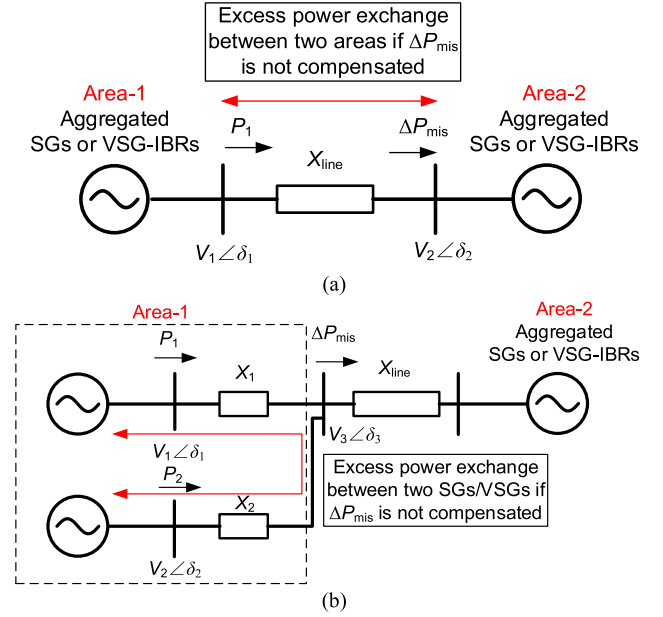


Fig. 2. Theoretical analysis of LFO instability mechanism: (a) inter-area mode, (b) local mode.

the reactive power output due to 2% droop may be beyond the reactive power capability. In this paper, the criterion is slightly modified as 30%Q/3%V (10% droop) to design:

$$D_{\text{qVSG}} = \frac{\Delta \overline{Q}}{\Delta \overline{V}_{\text{id}}} = \frac{30\%}{3\%} = 10 \quad (12)$$

which means a 3% terminal voltage change leads to 30% reactive power output.

The gain of the reactive power loop K corresponds to the exciter time constant T_a and shares the same unit as seconds. However, K should not inherit the value from T_a as discussed before. In this paper, K is designed as 0.46 s using the method of [26] by setting the cross-over frequency as 6 Hz and the phase margin as 120° .

III. THEORETICAL ANALYSIS OF LFO MECHANISM AND A SELECTION AND VALIDATION METHOD OF VSG-IBR SUBSTITUTION LOCATION SELECTION

A general instability mechanism of LFO is briefly analyzed based on a simplified two-area system, as shown in Fig. 2. After that, a selection and validation method of VSG-IBR substitution location is first proposed to further investigate the detailed stability mechanism of the inter-area oscillation modes induced by the substitution of VSG-IBRs for SGs. The proposed method is able to guide the selection of the VSG-IBR substitution locations. Moreover, it can assess the impact of VSG controls on damping of the concerned inter-area oscillation mode in different substitution locations.

A. Instability Mechanism Analysis of LFO

In Fig. 2(a), Area-1 and Area-2 are represented by aggregated SGs or VSG-IBRs. The power delivered from Area-1 to Area-2

can be calculated as:

$$\overline{P}_1 = \frac{\overline{V}_1 \overline{V}_2}{\overline{X}_{\text{line}}} \sin(\delta_1 - \delta_2) \approx \frac{\overline{V}_1 \overline{V}_2}{\overline{X}_{\text{line}}} (\delta_1 - \delta_2) \quad (13)$$

When there is a perturbation in Area-2 (such as a load variation), a power mismatch $\Delta \overline{P}_{\text{mis}}$ will appear with a sudden change of (virtual) rotor angle $\Delta \delta_2$:

$$\Delta \overline{P}_{\text{mis}} = \frac{\overline{V}_1 \overline{V}_2}{\overline{X}_{\text{line}}} (0 - \Delta \delta_2) \quad (14)$$

The Area-1 must respond to $\Delta \delta_2$ with $\Delta \delta_1$ to compensate for the power mismatch:

$$\Delta \overline{P}_1 = \frac{\overline{V}_1 \overline{V}_2}{\overline{X}_{\text{line}}} (\Delta \delta_1 - \Delta \delta_2) \rightarrow 0 \quad (15)$$

If the power mismatch $\Delta \overline{P}_{\text{mis}}$ cannot be compensated timely, the excess power will exchange between the two areas and cause inter-area oscillation [14]. The inter-area oscillation can be analyzed by eigenvalue-based stability assessment, and the inter-area mode reflects its damping and oscillation frequency. SG uses governor droop response and speed deviation damping to produce $\Delta \delta_1$ that eliminates the power mismatch. VSG uses the active power loop damping coefficient D_{pVSG} to provide similar functions. The VSG-IBR can provide faster response and more damping to the system through adjusting the control parameters in a wider range, as discussed in previous Section-C.

Inside Area-1, we assume two SGs (or VSG-IBRs) are connected in parallel, as shown in Fig. 2(b). Multiple parallel connected SGs (or VSG-IBRs) can be analyzed in a similar way. The perturbation outside Area-1 will be reflected at the terminal as $\Delta \overline{P}_{\text{mis}}$ and $\Delta \delta_3$. Then the power mismatch needs to be compensated by SG1 and SG2 (or VSG-IBR1 and VSG-IBR2):

$$\Delta \overline{P}_1 + \Delta \overline{P}_2 = \frac{\overline{V}_1 \overline{V}_3}{\overline{X}_1} (\Delta \delta_1 - \Delta \delta_3) + \frac{\overline{V}_2 \overline{V}_3}{\overline{X}_2} (\Delta \delta_2 - \Delta \delta_3) \rightarrow 0 \quad (16)$$

If the power mismatch $\Delta \overline{P}_{\text{mis}}$ cannot be eliminated due to different control effects and frequency response abilities, excess power will circulate between SG1 and SG2 (or VSG-IBR1 and VSG-IBR2), and cause power oscillation [10]. The power oscillation can also be analyzed by eigenvalue-based stability assessment and the local mode reflects its damping and oscillation frequency.

The inter-area oscillation reflected by inter-area modes (Fig. 2(a).) will be studied further by eigenvalue-based stability assessment and validated by time-domain simulations in later Sections IV and V.

B. Selection and Validation of VSG-IBR Substitution Location

The selection and validation procedure of VSG-IBR substitution location is shown in Fig. 3.

Before VSG-IBR substitution, eigenvalue-based analysis needs to be conducted on the original system (all SGs). Then the inter-area modes can be identified if the highest 5 PFs are from SGs in different areas. The inter-area oscillation modes could also be observed from time-domain simulations (introduced in later Section V-A). In a small-scale system (like the 2A4M

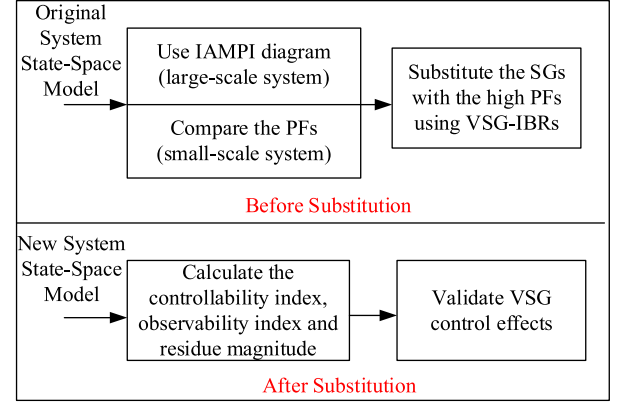


Fig. 3. Selection and validation procedure of VSG-IBR substitution location.

system in Section IV), it is easy to identify which groups the SGs participate in. However, in a large-scale system (like the 16M68B system in Section V), every inter-area mode involves SGs in different areas, and every SG also participates in different inter-area modes, which makes analysis complicated. The directions of SG right eigenvectors can be used to observe the directions of SG participation [33], which are called mode shapes in some studies [16]. SGs with the same right eigenvector directions are considered a group in an inter-area mode, and SGs with different right eigenvector directions can reflect a mode.

However, the mode shape cannot reflect the extent of SG participation, and PF cannot reflect the directions of SG participation. Hence, this paper combines the concepts of mode shape and PF, and proposes an inter-area mode participation index (IAMPI):

$$\text{IAMPI}^{(k)} = PF_i^{(k)} \cdot \frac{\varphi_i^{(k)}}{|\varphi_i|^{(k)}} = \frac{\varphi_i^{(k)} \cdot \psi_i^{(k)}}{\max(\varphi_i^{(k)} \cdot \psi_i^{(k)})} \cdot \frac{\varphi_i^{(k)}}{|\varphi_i|^{(k)}} \quad (17)$$

where $\varphi_i^{(k)}$ and $\psi_i^{(k)}$ are the right and left eigenvectors of the i th SG to the k th mode. The IAMPI takes the product of relative PF (PF/PF_{max}) and the normalized right eigenvector. Therefore, the direction of the IAMPI vector is the same as the right eigenvector direction, which indicates the participation direction of SGs in a mode (mode shape). The length of the IAMPI vector indicates how largely SGs influence a mode (PF).

Hence, the selection of substitution locations could refer to the IAMPI vector. In this paper, the substitution location is equivalent to the participation factor of the SG which will be substituted. The SGs in different locations that will be substituted by VSG-IBRs may have different PFs on the inter-area oscillation modes. Thus, these VSG-IBRs will show different damping characteristics after substituting those SGs. Furthermore, the impact of VSG-IBR key control parameters on inter-area oscillations can also vary in different substitution locations.

After VSG-IBR substitution, controllability, observability, and residues are calculated to assess the impact of VSG control on system damping in different substitution locations, as the VSG can be considered as a damping control to the entire system.

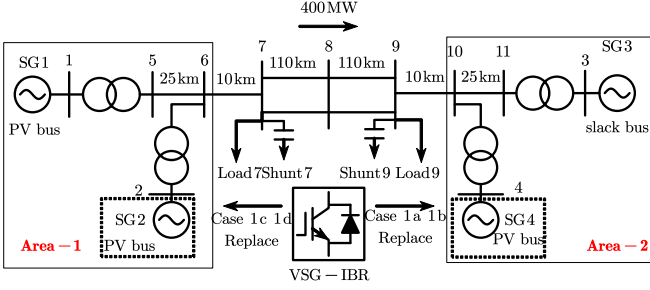


Fig. 4. Configurations of the 2A4M system.

The output power of VSG (P) is selected as the system control input $\mathbf{u}$, and its controllability represents the control effects of VSG on the mode. The virtual rotor angle of VSG (δ_{VSG}) is selected as the system output $\mathbf{y}$, and its observability represents the how well it can reflect the mode. Hence, the controllability index can be calculated as the magnitude of the product of the left eigenvector and the matrix $\mathbf{B}$ of the state-space model. The observability index can be calculated as the magnitude of the product of the matrix $\mathbf{C}$ and the right eigenvector. Residue of mode k ($\mathbf{R}^{(k)}$) is the product of the observability ($\mathbf{C}_{j \times i} \cdot \varphi_{i \times 1}^{(k)}$) and controllability ($\psi_{1 \times i}^{(k)} \cdot \mathbf{B}_{i \times u}$), which reflects the control effects of control input $\mathbf{u}$ (the uth VSG power) on system output $\mathbf{y}$ (the yth SG or VSG rotor angle):

$$\mathbf{R}^{(k)}_{y \times u} = \mathbf{C}_{y \times i} \cdot \varphi_{i \times 1}^{(k)} \cdot \psi_{1 \times i}^{(k)} \cdot \mathbf{B}_{i \times u} \quad (18)$$

From (18), the controllability, observability and residues could be obtained and used to validate the effectiveness of substitution locations based on IAMPI diagram. The detailed procedure through employing the proposed method will be applied in later Sections IV and V.

IV. IMPACT ANALYSIS OF THE VSG-IBR SUBSTITUTION LOCATIONS AND CONTROL PARAMETERS ON INTER-AREA OSCILLATIONS

A. Inter-Area Oscillation Analysis of the 2A4M System

Kundur's 2A4M system is first employed, as SG parameters in this system are identical, and the system is symmetrical to the area interface. The system diagram is shown in Fig. 4 with the system parameters listed in [4]. During the state-space modeling process, the transformers and lines are represented with equivalent impedances (π -equivalent circuit for lines and T-equivalent circuit for transformers) in the considered test systems (2A4M and 16M68B systems). Hence, the effects of VSG-IBR substitution are obvious to analyze the impacts of substitution locations and control parameters.

The eigenvalue analysis is based on the state-space models in Section II, and the results are shown in Table I. The inter-area mode is unstable with 0.58 Hz frequency and -0.99% damping. The PFs of the four machines are sorted as $SG3 > SG4 > SG1 > SG2$. In time domain simulation, the inter-area oscillation is triggered by a 5% step change of field voltage reference in SG1 at $t = 1$ s, and this step change lasts 0.2 s. Rotor speeds of the four SGs in Fig. 5 show that the oscillation

TABLE I
EIGENVALUE ANALYSIS OF THE 2A4M SYSTEM

Frequency and damping of the inter-area mode	
Freq (Hz)	Damping (%)
0.58	-0.99
Highest four participation factors (PF/PF _{max})	
$\omega_3, (\delta_{r3}), \omega_4, (\delta_{r4}), \omega_1, (\delta_{r1}), \omega_2, (\delta_{r2})$	
$SG3 (1.00) > SG4 (0.80) > SG1 (0.59) > SG2 (0.37)$	

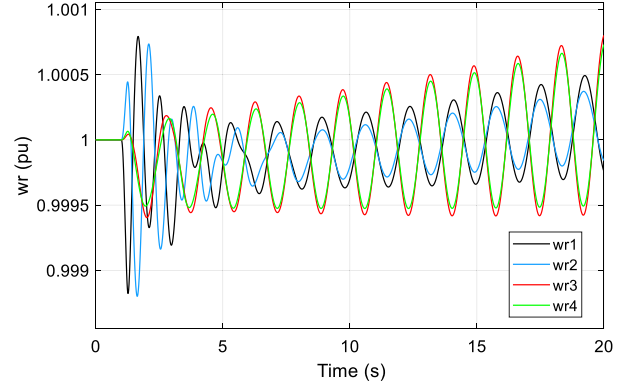


Fig. 5. Rotor speeds of the four SGs in the 2A4M system.

TABLE II
IMPACTS OF SYSTEM AND SG PARAMETERS ON PFs

PFs of SG Rotor Speeds	
Parameter 1 Reverse power flow direction	Parameter 2 Line length
$SG1 (1.00) > SG2 (0.80)$ $> SG3 (0.53) > SG4 (0.33)$	$SG3 (1.00) = SG4 (1.00)$ $> SG1 (0.16) = SG2 (0.16)$
Parameter 3 SG output power	Parameter 4 SG Inertia
$SG3 (1.00) > SG4 (0.80)$ $> SG1 (0.65) > SG2 (0.33)$	$SG3 (1.00) > SG4 (0.82)$ $> SG2 (0.45) > SG1 (0.19)$

magnitudes of the four SGs match the order of PFs. Hence, **Finding 1 is concluded that the SGs with higher PFs have more severe oscillations.**

The impacts of the system and SG parameters (substitution locations) on PF are further investigated. The direction of power flow and line lengths are tested as system factors. The output power and inertia are tested as SG factors. The analysis results are shown in Table II.

In the original system, 400 MW is delivered from Area-1 to Area-2. Then exchange the load and shunt capacitances of Bus-7 and Bus-9 so that 400 MW is delivered from Area-2 to Area-1. The PFs of the four machines are sorted as $SG1 > SG2 > SG3 > SG4$ (see Table II Parameter 1). Therefore, SGs in the receiving end of the power system have higher PFs.

Adding 25 km lines between Bus-2 & Bus-6 and between Bus-4 & Bus-11 makes the network completely symmetrical. PFs of $SG3$ & $SG4$ become identical, and PFs of $SG1$ & $SG2$ also become identical, as shown in Table II Parameter 2. Hence, SGs with the same line lengths to the area interface have identical PFs. On the other hand, it is also validated that the SGs connected by longer lines to the inter-area oscillation interface have higher

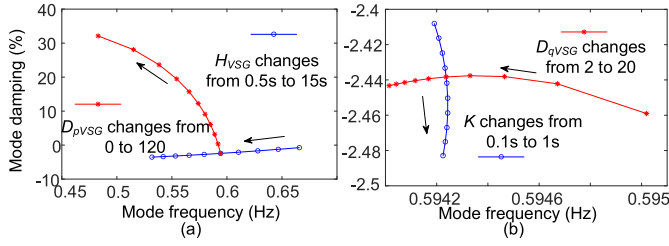


Fig. 6. Impacts of VSG parameters on inter-area modes: (a) H_{VSG} and D_{pVSG} , (b) K and D_{qVSG} .

PFs, because the PF of SG2 increases to the same value as the PF of SG1 after adding the extra line. This finding matches well with [9] that less kinetic energy of SGs far from the inter-area oscillation interface can be absorbed by the network which leads to slowly damped oscillations, and thus higher PFs in the inter-area modes.

When SG2 increases 200 MW output power and SG1 decreases 200 MW output power, the PF of SG2 decreases while the PF of SG1 increases, as shown in Table II Factor 3. Therefore, SGs with smaller output power have higher PFs. When SG2 doubles its inertia, the PF of SG2 exceeds the PF of SG1, as shown in Table II Parameter 4. Hence, SGs with larger inertia have higher PFs. The impacts of the system and SG parameters on PF are summarized as **Finding 2**:

- 1) SGs in the receiving end of the power system (the power flows into this area due to a smaller generation capacity compared to larger loads) have higher PFs.
- 2) SGs connected by long lines to the area interface have higher PFs.
- 3) SGs with smaller output power have higher PFs.
- 4) SGs with larger inertia have higher PFs.

B. Impact Analysis of VSG-IBR Control Parameters With Fixed Substitution Location

The original VSG-IBR parameters H_{VSG} , D_{pVSG} , K and D_{qVSG} are 6.5 s, 60, 0.46 s and 10, respectively. Impacts of the control parameters are studied under the scenario where VSG-IBR substitutes SG4, and the results are shown in Fig. 6. As D_{pVSG} changes from 0 to 120, the mode frequency decreases, and damping increases significantly, as shown in Fig. 6(a). Consequently, it is the main parameter for VSG-IBR to provide damping to the system. As H_{VSG} changes from 0.5s to 15s, both the mode frequency and damping decrease, as shown in Fig. 6(a). But the impact of H_{VSG} on mode damping is less significant than D_{pVSG} .

As shown in Fig. 6(b), two reactive power loop parameters have little influence on inter-area mode. Hence, **Finding 3** can be revealed: **Impacts of H_{VSG} and D_{pVSG} are more significant than K and D_{qVSG} , and D_{pVSG} is crucial for VSG-IBR to provide damping to the system.**

C. Impact Analysis of the VSG-IBR Substitution Locations

In this section, the SG in the 2A4M system will be substituted by the VSG-IBR with the same power rating and inertia (the

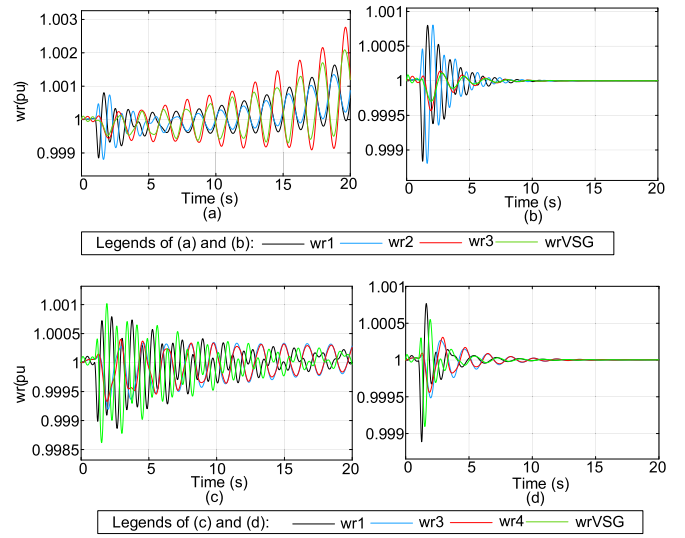


Fig. 7. Simulation results of case 1: Fig. (a)–(d) correspond with C1-1 to C1-4.

TABLE III
EIGENVALUE ANALYSIS OF CASE 1

Frequency and damping of inter-area modes			
C1-1		C1-2	
Freq (Hz)	Damping (%)	Freq (Hz)	Damping (%)
0.59	-2.44	0.57	12.27
C1-3		C1-4	
Freq (Hz)	Damping (%)	Freq (Hz)	Damping (%)
0.56	1.06	0.55	9.98

VSG-IBR is assumed as an aggregated wind park/solar park or a large-scale VSC-HVDC station). Two main factors are considered: PFs of the substituted SGs and the VSG-IBRs with proper design and without the design of D_{pVSG} .

For the first factor, SG2 and SG4 are chosen to be substituted. SG2 is in the power flow sending area, and SG4 is in the power flow receiving end, and these two SGs have the same parameters and are connected by lines with the same length. According to the analysis in Section-A and B, the PF of SG4 is higher than the PF of SG2 only because of the power flow direction. Due to the symmetrical property of the system, the scenario of choosing SG1 and SG3 is neglected because the results are similar to choosing SG2 and SG4.

The second factor will focus on the impact of D_{pVSG} when VSG-IBRs substitute SGs with different PFs, and the other three parameters are tuned based on Section-C. Hence, the Case-1 (C1) is constructed as follows:

- C1-1 and C1-2: VSG-IBR substitutes SG4 with $D_{pVSG} = 0$ (inherited from SG speed deviation damping D_{pSG}) and designed value $D_{pVSG} = 60$, respectively.
- C1-3 and C1-4: VSG-IBR substitutes SG2 with $D_{pVSG} = 0$ and designed value $D_{pVSG} = 60$, respectively.

The simulation and eigenvalue analysis results are shown in Fig. 7 and Table III, respectively.

Comparing the results of C1-1 with the original results in Table I, the damping decreases from -0.99% to -2.44% after the VSG-IBR substitutes SG4, as shown in Table III C1-1.

TABLE IV
CONTROLLABILITY INDICES, OBSERVABILITY INDICES AND RESIDUE
MAGNITUDES OF C1

Case	Controllability Index	Observability Index	Residue Magnitude
C1-1	1.98	0.0044	0.0088
C1-2	2.45	0.0042	0.0104
C1-3	1.99	0.0027	0.0053
C1-4	2.24	0.0025	0.0056

Comparing the results of C1-3 with the original results in Table I, the damping increases from -0.99% to 1.06% after the VSG-IBR substitutes SG2, as shown in Table III C1-3. The undamped oscillations in Fig. 7(a) and the damped oscillations in Fig. 7(c) also verify the eigenvalue analysis. Hence, **Finding 4** can be obtained that **the VSG-IBR without the design of D_{pVSG} (inherited from SG speed deviation damping D_{pSG}) is recommended to substitute the SG with a lower PF (i.e., SG2) to decrease the negative damping provided to the system. The VSG-IBR reduces the damping significantly when it substitutes the SG with a higher PF (i.e., SG4).**

Comparing the results of C1-2 with C1-1, the damping increases from -2.44% to 12.27% when the D_{pVSG} is properly designed, as shown in Table III C1-2. Comparing the results of C1-4 with C1-3, the damping increases from 1.06% to 9.98% when the D_{pVSG} is properly designed, as shown in Table III C1-4. The fast-damped oscillations in Fig. 7(b) and the slowly damped oscillations in Fig. 7(d) also validate the eigenvalue analysis. Therefore, **Finding 5** can be achieved that **the VSG-IBR with properly designed D_{pVSG} is recommended to substitute the SG with a higher PF (i.e., SG4).** In this way, the VSG-IBR can contribute more positive damping to the system through tuning D_{pVSG} .

D. Validation of VSG-IBR Substitution Location

The controllability indices, observability indices and the residue magnitudes of C1 are listed in Table IV. Comparing C1-1 and C1-3, the controllability indices are similar, but the observability of C1-1 is better. Hence, the VSG-IBR substituting SG4 better represents the inter-area mode. Comparing C1-2 and C1-4 after tuning D_{pVSG} , all three indices have significantly improved, and are larger in C1-2, indicating that the VSG-IBR substituting SG4 has better control effects. Hence, the findings in previous Section IV-C are validated.

V. IMPACT ANALYSIS OF VSG-IBR CENTRALIZED AND DISTRIBUTED SUBSTITUTION STRATEGIES ON INTER-AREA OSCILLATIONS

This section will discuss the impact of VSG-IBR centralized and distributed substitution strategies on inter-area oscillations.

A. Inter-Area Oscillation Analysis of the 16M68B System

The New York New England 16M68B system is an extended version of the New England 10M39B system [37]. The system diagram is shown in Fig. 8. Five areas are separated in this system: New England (NE), New York (NY), Area-3, Area-4,

TABLE V
EIGENVALUE AND PF ANALYSIS OF 16M68B SYSTEM

Frequency and damping of inter-area modes			
Modes	Freq (Hz)	Damping (%)	Participation Areas
Mode 1	0.43	-1.5	SG14, SG15, SG13, SG6, SG9 (A3-A4-NY-NE)
Mode 2	0.64	0.19	SG14, SG16, SG15, SG13, SG12 (A3-A5-A4-NY)
Mode 3	0.73	-0.15	SG13, SG12, SG15, SG14, SG6 (NY-A4-A3-NE)
Mode 4	0.88	2.74	SG16, SG15, SG12, SG13, SG14 (A5-A4-NY-A3)
Mode 5	1.19	3.52	SG12, SG10, SG13, SG1, SG8 (NY-NE)
Mode 6	1.24	3.44	SG13, SG1, SG12, SG10, SG8 (NY-NE)

TABLE VI
SGS AND AREAS IN TWO PARTICIPATION GROUPS OF INTER-AREA MODES

Groups	Group 1	Group2
Mode 1	SG1 to SG13 (NE, NY)	SG14, SG15, SG16 (A3, A4, A5)
Mode 2	SG1 to SG9, SG14 (NE, A3)	SG10 to SG13, SG15, SG16 (NY, A4, A5)
Mode 3	SG1 to SG9, SG15, SG16 (NE, A4, A5)	SG10 to SG13, SG14 (NY A3)

and Area-5 (A3, A4&A5). SG13, SG14, SG15, and SG16 are aggregated SGs, and SG16 is the slack bus. Red arrow lines mark the direction of power flows in the interfaces of coherent areas. The basic parameters of the system can be found in [33]. The inter-area oscillation is triggered by a 2% step in the field voltage reference of SG3, which starts at $t = 1s$ and lasts for 0.5s. The eigenvalue analysis results are shown in Table V.

Six modes (110 modes in all) are classified as inter-area modes because the top five PFs are from SGs in different areas. The first three inter-area modes are under discussion in the following sections as the other three modes remain stable after VSG-IBR integration. The IAMPI diagram of the first three inter-area modes is shown in Fig. 9, which demonstrates the mode damping and two opposite participation directions of SGs. SGs and their belonged areas are separated into two opposite participation groups in different inter-area modes, as shown in Table VI.

Then SG1 and SG15 can be chosen to mainly reflect mode 1 and mode 2 by calculating the rotor angle difference, because they are in different groups of mode 1 and mode 2, but the same group of mode 3, as shown in Table VI. Similarly, the rotor angle difference of SG14 and SG15 mainly reflects mode 2 and mode 3, and the rotor angle difference between SG13 and SG15 mainly reflects mode 1 and mode 3. Simulations of rotor speeds and rotor angle differences of SG1&SG15, SG14&SG15, and SG13&SG15 are shown in Fig. 10. All rotor speed waveforms display significant phase differences. Especially for SG1&SG15 and SG13&SG15, the rotor speeds have nearly 180° phase differences, so the two SGs show alternative acceleration and deceleration. The rotor angle difference waveforms contain

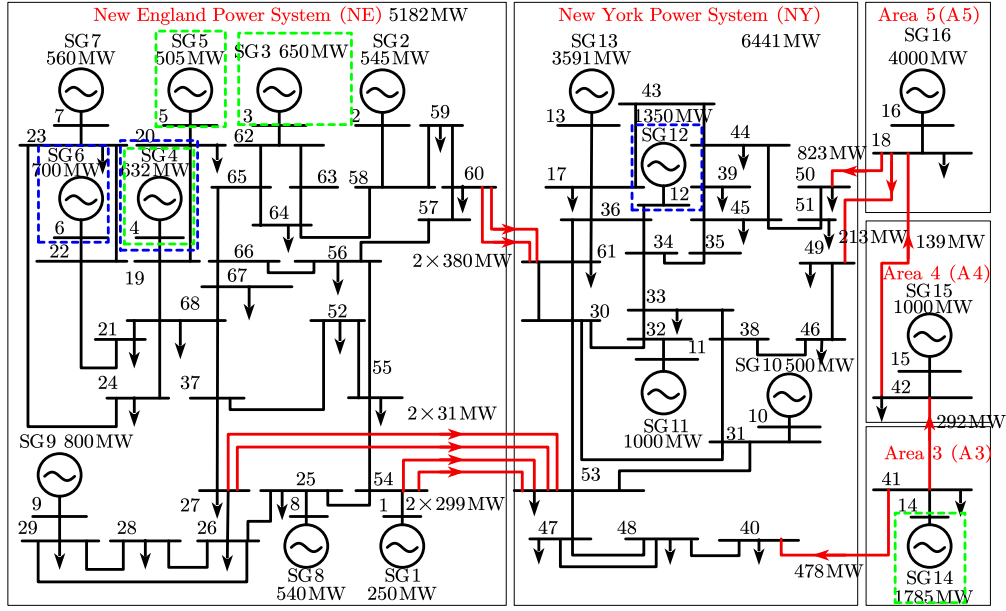


Fig. 8. Line diagram of the 16M68B system: the substituted SGs in test case 2 (SG4, SG6 and SG12) are encircled by blue dotted lines, and the substituted SGs in test case 3 (SG3, SG4, SG5 and SG14) are encircled by green dotted lines.

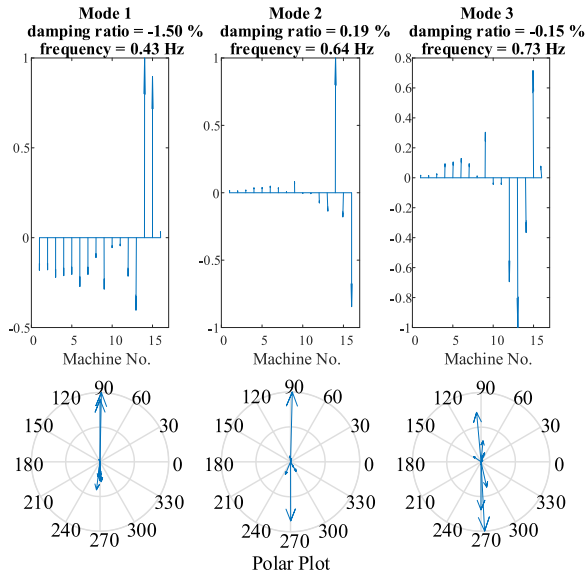


Fig. 9. IAMPI diagram of the original 16M68B system.

mode frequencies, which can be demonstrated by Prony or Fast Fourier Transformation (FFT) analysis.

B. Impact Analysis of VSG-IBR Centralized and Distributed Substitution Strategies

Case-2 (C2) and Case-3 (C3) are designed as follows.

- C2-1 and C2-2: One VSG-IBR substitutes SG12 with D_{pVSG} as 2 and 60, respectively. (The value of 2 is inherited from the SG speed deviation damping D_{pSG})
- C2-3 and C2-4: Two VSG-IBRs substitute SG4 and SG6 with D_{pVSG} as 2 and 60, respectively. The output power of

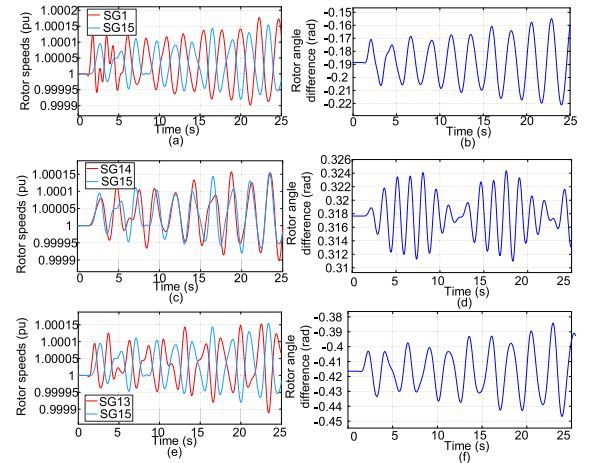


Fig. 10. Simulations of the original 16M68B system: (a) and (b) rotor speeds and of rotor angle difference of SG1 and SG15, (c) and (d) rotor speeds and of rotor angle difference of SG14 and SG15, (e) and (f) rotor speeds and of rotor angle difference of SG13 and SG15.

SG12 is nearly the same as the sum output power of SG4 and SG6 ($1350 \text{ MW} \approx 700 + 632 \text{ MW}$). C2 is marked as blue dotted boxes in Fig. 8.

- C3-1 and C3-2: One VSG-IBR substitutes SG14 with D_{pVSG} as 2 and 60, respectively.
- C3-3 and C3-4: Three VSG-IBRs substitute SG3, SG4 and SG5 with D_{pVSG} as 2 and 60, respectively. The output power of SG14 is nearly the same as the sum output power of SG3, SG4 and SG5 ($1785 \text{ MW} \approx 650 \text{ MW} + 632 \text{ MW} + 505 \text{ MW}$). C3 is marked as green dotted boxes in Fig. 8.

K and D_{qVSG} are designed in the same way as the 2A4M system, H_{VSG} inherits the value from H_{SG} of the substituted SG.

TABLE VII
EIGENVALUE ANALYSIS OF CASE 2

Frequency and damping of inter-area modes				
Cases	C2-1		C2-2	
Modes	Freq (Hz)	Damping (%)	Freq (Hz)	Damping (%)
Mode1	0.43	-1.63	0.43	0.40
Mode2	0.63	0.30	0.63	0.79
Mode3	0.74	-0.54	0.74	3.02
Cases	C2-3		C2-4	
Modes	Freq (Hz)	Damping (%)	Freq (Hz)	Damping (%)
Mode1	0.44	0.67	0.44	9.41
Mode2	0.64	0.89	0.634	2.48
Mode3	0.74	0.79	0.75	3.27

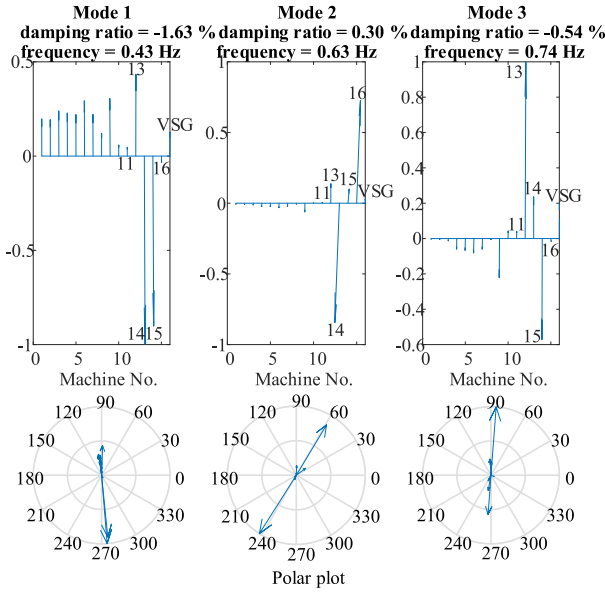


Fig. 11. IAMPI diagram of the 16M68B System in C2-1.

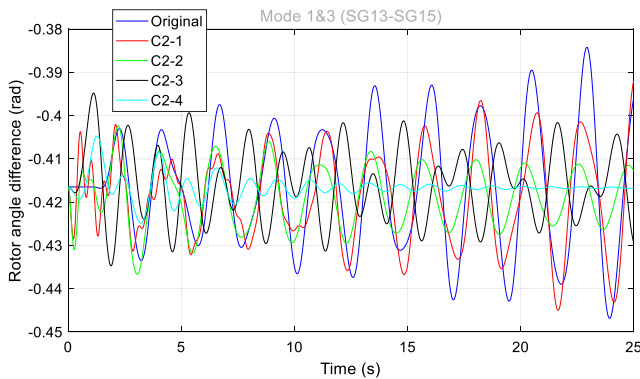


Fig. 12. Test case C2: rotor angle differences between SG13 and SG15.

1) *Test Case-2*: The eigenvalue analysis, IAMPI diagram and the time-domain simulation results of C2 are shown in Table VII, Figs. 11, and 12, respectively. The IAMPI diagram shown in Fig. 11 displays C2-1 after VSG-IBR substitutes SG12. The VSG-IBR remains the same participation group as SG12

in three modes. Besides, the participation groups of other SGs also remain unchanged although the directions of two groups are reverse by 180° , when comparing Figs. 9 and 11. Hence, the original IAMPI diagram can be utilized in other cases. The simulation results only demonstrate the rotor angle difference between SG13 and SG15 as an example.

Comparing the results of C2-1 with the original results in Table V, the damping of mode 1 and mode 3 decreases a little (from -1.50% to -1.63% , from -0.15% to -0.54% , respectively), and the damping of mode 2 increases a little (from 0.19% to 0.30%) after the VSG-IBR substitutes SG12, as shown in Table VII C2-1. In general, the system stability slightly increases, verified by a slightly lower oscillation magnitude of the red line than the original blue line, as shown in Fig. 12.

Comparing the results of C2-3 with C2-1, the damping of all three modes is improved after two VSG-IBRs substitute SG4 and SG6, as shown in Table VII C2-3. The damped oscillations of the black line in Fig. 12 verify the eigenvalue analysis. Hence, **Finding 6** is concluded that **VSG-IBRs without the design of D_{pVSG} (inherited from SG speed deviation damping D_{pSG}) are recommended to substitute SGs in a distributed rather than a centralized strategy. The negative damping effects of the VSG-IBR without the design of D_{pVSG} might be dispersed by distributed substitution.**

Comparing the results of C2-2 with C2-1, the damping of all three modes is significantly improved when the D_{pVSG} is properly designed, as shown in Table VII C2-2. Among all three modes, the damping of mode 3 has the most significant increase (from -0.54% to 3.02%), because SG12 has the highest PF of mode 3 (0.7) compared to the PFs of the other two modes (0.21 and 0.06, respectively), as shown in Fig. 9. Hence, it reflects **Finding 5** in large-scale power system scenario. **The properly designed D_{pVSG} improves the damping significantly in the modes of which the substituted SGs have higher PFs.** The system stability is generally improved as the green line shows damped oscillations in Fig. 12.

Comparing the results of C2-4 with C2-3, the damping of all three modes is significantly improved when the D_{pVSG} is properly designed, as shown in Table VII C2-4. Among all three modes, the damping of mode 1 has the most significant increase (from 0.67% to 9.41%), because the PFs of SG4 and SG6 are the highest in mode 1 (SG4: 0.2, SG6: 0.27, respectively) compared to the other two modes (SG4: 0.04 and 0.09, SG6: 0.05 and 0.13, respectively), as shown in Fig. 9. On the other hand, the effects of C2-4 are much better than centralized substitute strategy of C2-2. Hence, the **Finding 6** can extend to VSG-IBRs with properly designed D_{pVSG} . **The positive damping effects of the VSG-IBR with properly designed D_{pVSG} might contribute to more modes by distributed substitution.** The system stability is generally significantly improved as the cyan line shows the fastest damped oscillations in Fig. 12.

2) *Test Case-3*: The eigenvalue analysis and time-domain simulation results of C3 are shown in Table VIII and Fig. 13, respectively.

Comparing the results of C3-1 with the original results, the damping of all three modes decreases after the VSG-IBR substitutes SG14, as shown in Table VIII C3-1. Among all three

TABLE VIII
EIGENVALUE ANALYSIS OF CASE 3

Frequency and damping of inter-area modes				
Cases	C3-1		C3-2	
Para- meters	Freq (Hz)	Damping (%)	Freq (Hz)	Damping (%)
Mode1	0.43	-1.61	0.43	6.69
Mode2	0.63	-0.63	0.62	8.57
Mode3	0.73	-0.35	0.72	0.38
Cases	C3-2		C3-4	
Para- meters	Freq (Hz)	Damping (%)	Freq (Hz)	Damping (%)
Mode1	0.45	0.96	0.45	10.22
Mode2	0.64	0.94	0.64	2.40
Mode3	0.74	0.52	0.75	2.35

TABLE IX
RESIDUE MAGNITUDES OF C2 AND C3

Modes	VSG Reside Magnitudes			
	VSG12	VSG4	VSG6	
C2-1&3 Mode1	0.479	0.819	0.786	
C2-2&4 Mode1	0.479	0.809	0.776	
C2-1&3 Mode2	0.366	0.467	0.456	
C2-2&4 Mode2	0.392	0.452	0.440	
C2-1&3 Mode3	0.718	0.470	0.464	
C2-2&4 Mode3	0.716	0.447	0.441	
Modes	VSG14	VSG3	VSG4	VSG5
C3-1&3 Mode1	0.671	0.697	0.763	0.721
C3-2&4 Mode1	0.718	0.695	0.756	0.716
C3-1&3 Mode2	0.873	0.290	0.443	0.428
C3-2&4 Mode2	0.943	0.289	0.422	0.407
C3-1&3 Mode3	0.321	0.191	0.426	0.418
C3-2&4 Mode3	0.244	0.172	0.363	0.355

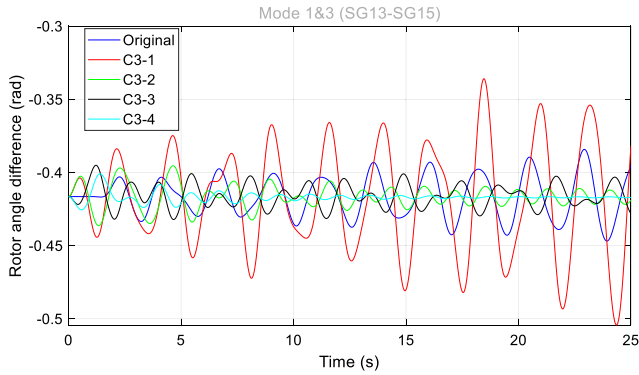


Fig. 13. Test case C3: rotor angle differences between SG13 and SG15.

modes, the damping of mode 2 has the most significant decrease (from 0.19% to -0.63%). The improperly designed D_{pVSG} has the most significant effects on the damping decrease of mode 2, because SG14 has the highest PF of mode 2 (1.0), as shown in Fig. 9. Hence, it also reflects **Finding 4** in the large-scale power system scenario. **The VSG-IBR without the design of D_{pVSG} (inherited from SG speed deviation damping D_{pSG}) worsens the damping significantly in the modes of which the substituted SGs have higher PFs.** In general, the system stability deteriorates as the red line's undamped oscillations show a larger oscillation magnitude than the original blue line in Fig. 13.

Distributed substitute of VSG-IBRs improves the damping of all three modes compared to the centralized substitute in C3-1, as shown in Table VIII C3-3. The damped oscillations of the black line in Fig. 13 verify the analysis. The properly designed D_{pVSG} value improves the system damping both in centralized or distributed substitute scenarios, as shown in Table VIII C3-2 and C3-4, respectively. Compare the results of C3-2 with C3-4, damping has the most significant increase in mode 2, because the SG14 has a significantly higher PF (1.0) than the PFs of SG3 – SG5 in mode 2 (SG3: 0.02, SG4: 0.04, SG5: 0.04, respectively), as shown in Fig. 9. However, the distributed substitute of C3-4 provides more positive damping to mode 1 and mode 3 than the centralized substitute, which is due to the higher PFs of SG3 –SG5 in mode 1 and mode 3. In general, C3-4 makes the

system more stable, because the cyan line shows faster damped oscillations than the green line, as seen in Fig. 13.

C. Validation of VSG-IBR Substitution Strategy

As the residue magnitudes of three modes in C2 and C3 are too small, they are divided by the maximum residue magnitude of the corresponding control input $\mathbf{u}$, which are listed in Table IX. The controllability indices and observability indices are ignored for simplicity as the residue magnitude can demonstrate similar information. It can be seen that the sums of residue magnitudes (VSG4+VSG6 in C2, VSG3+VSG4+VSG5 in C3) are larger than residue magnitudes of a single VSG (VSG12 in C2, VSG14 in C3). Hence, the finding in previous Section V-B that better control effects of distributed substitution than centralized distribution has been validated. Besides, the residues also show that the VSG12 has limited control effect over mode2, and the VSG 14 has limited control effect over mode3, which validates the finding in Tables VII C2-2 and VIII C3-2.

It is worth mentioning that tuning control parameters cannot stabilize the system in all scenarios, or the control parameters could only be adjusted within a certain range. Under these circumstances, supplementary damping controllers (SDCs) should be implemented in VSG-IBRs for desired system damping/stability margin.

VI. CONCLUSION

This paper conducts a comprehensive analysis about the impacts of VSG-IBRs integration on power system inter-area oscillations. It is the first trial to consider the impact of substitution locations and substitution strategies on inter-area oscillations. The main findings are listed below:

- 1) The substitution location determines the participation factors (PFs) of SGs. SGs with higher PFs on the inter-area modes experience more severe oscillations. SGs in the receiving end of the power system, connected by longer lines, with larger inertia and smaller output power have higher PFs. Thus, VSG-IBRs will show distinct damping characteristics after substituting these SGs. Furthermore, the impact of VSG-IBR key control

parameter D_{pVSG} on inter-area oscillations can also vary in different substitution locations.

- 2) A selection and validation procedure of VSG-IBR substitution location is proposed. The PF and Inter-area mode participation index (IAMPI) diagram can help select the SGs to be substituted. Then the controllability index, observability index and residues are calculated to validate the control effects of VSG-IBRs after substituting the selected SGs.
- 3) VSG-IBRs without design of D_{pVSG} (inherited from SG speed deviation damping D_{pSG}) are recommended to substitute SGs with lower PFs to decrease the negative damping provided to the inter-area modes. In a large-scale system, the distributed substitution strategy (i.e., distributed VSG-IBRs in different locations) disperses the negative damping effects, so it has less negative impacts on the system stability than centralized substitution strategy (e.g., a large HVDC station operating in VSG control).
- 4) VSG-IBRs with properly designed D_{pVSG} are recommended to substitute SGs with higher PFs to provide more positive damping to the inter-area modes. In a large-scale system, the distributed substitution strategy contributes the positive damping to more modes to further improve the system stability than the centralized substitution strategy.

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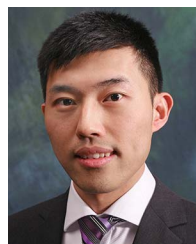


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